



AI prediction of extracorporeal shock wave lithotripsy outcomes for ureteral stones by machine learning-based analysis with a variety of stone and patient characteristics

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Abstract

We propose an artificial intelligence prediction method for extracorporeal shock wave lithotripsy treatment outcomes by analysis of a wide variety of variables. We retrospectively reviewed the records of 171 patients from between January 2009 and November 2019 that underwent shock wave lithotripsy at Wakayama Medical University, Japan, for ureteral stones shown on preoperative non-contrast computed tomography. This prediction method consisted of stone area extraction, stone analyzing factor extraction from non-contrast computed tomography images, and shock wave lithotripsy treatment result prediction by a non-linear support vector machine for analysis of 15 input and automatic measurement factors. Input factors included patient age, skin-to-stone distance, and maximum ureteral wall thickness, and the automatic measurement factors included 11 non-contrast computed tomography image texture factors in the stone area and stone volume. Permutation feature importance was also applied to the artificial intelligence prediction results to analyze the importance of each factor relating to estimate decision grounds. The prediction performance was evaluated by five-fold cross-validation, it obtained 0.742 of the mean area under the receiver operating characteristic curve. The proposed method is shown by these results to have robust data diversity and effective clinical application. As a result of permutation feature importance, some factors that showed high *p*-values in the significant difference tests were thought to have a high contribution to the proposed prediction method. Future issues include validation using a larger volume of high-resolution clinical non-contrast computed tomography image data and the application of deep learning.

Keywords Ureteral stone · Shock wave lithotripsy · Support vector machine · Permutation feature importance · Artificial intelligence

Abbreviations

AI Artificial intelligence
AUC Area under the ROC curve
CV Cross-validation

GCFs Gradient concentration factors
IQR Interquartile range
MSD Mean stone density
NCCT Non-contrast computed tomography
PCNL Percutaneous nephrolithotomy
PFI Permutation feature importance
ROC Receiver-operating characteristic
SD Standard deviation
SSD Skin-to-stone distance
SVM Support vector machine
SWL Shock wave lithotripsy
URS Ureteroscopy
UWT Ureteral wall thickness
UWV Ureteral wall volume
VCSD Variation coefficient of stone density

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Introduction

Ureteroscopy (URS) and percutaneous nephrolithotomy (PCNL) have been performed more frequently for upper urinary tract calculi with the development of endoscopic technology. Extracorporeal shock wave lithotripsy (SWL) nonetheless remains a first-line treatment option because of its low invasiveness and safety, but it has a lower treatment success rate than URS or PCNL [1, 2]. Predicting SWL treatment outcomes is therefore important to ascertain the optimal treatment of patients with upper urinary tract stones.

Various factors measured by non-contrast computed tomography (NCCT) are useful in the prediction of SWL treatment outcomes [3–8]. Guidelines by the European Association of Urology and the American Urological Association both suggest treatment selection for upper urinary tract calculi mainly based on stone location and size [9, 10]. Other factors, such as skin-to-stone distance (SSD) and mean stone density (MSD), are also listed in these guidelines as influencing the success rate of SWL treatment. Recently, stone heterogeneity index and variation coefficient of stone density (VCSD) have been considered to be indicators of heterogeneity of CT attenuation values of stones. Ureteral wall thickness (UWT) and ureteral wall volume (UWV) have been considered as indicators of ureteral wall thickening in ureteral stones. Both have attracted attention as predictors of SWL treatment outcomes, but further investigation is required [6–8]. As described above, in addition to stone location and size, NCCT can provide a great deal of information that can be useful for predicting SWL treatment outcomes. However, due to errors of each predictor caused by differences in measurement methods and the complexity of systematic understanding, these predictors have not been effectively utilized in the selection of treatment methods. In the current study, we therefore focused on the automatic measurement of each predictor and the construction of algorithms by artificial intelligence (AI) to identify the outcome of SWL treatment.

Prediction of SWL treatment outcome by AI analysis using multiple factors has also been evaluated. Mannil et al. studied prediction of SWL treatment outcomes using various AI algorithms with a total of 224 three-dimensional image textures, patients' body mass index, stone size, and skin-to-stone distance (SSD) [11]. Prediction by random forest classifier with the selection of only a few texture parameters and the stone size led to a 0.81 area under the ROC curve (AUC).

We propose an AI-based method for predicting SWL treatment outcomes by machine learning with a wide variety of factors, such as CT image texture parameters on the target stone, the stone location and size, the environment

around the stone, and other patient characteristic factors. This proposed method is specifically designed to predict the outcome of first SWL treatment. It was retrospectively validated using multiple patients' data, including CT images of various resolutions and slice thicknesses taken with scanners from different manufacturers. Study using a dataset with inconsistent specifications is exceptionally challenging, but it is thought to be highly applicable to actual clinical practices. To analyze the importance of each factor in the prediction, the proposed method using multiple prediction factors was also applied to the permutation feature importance (PFI) [12].

Methods

Patients

We retrospectively reviewed the records of the 216 consecutive patients that underwent SWL at Wakayama Medical University for ureteral stones shown on preoperative NCCT between January 2009 and November 2019. The CT slice resolution ranged from 0.60×0.60 to 0.86×0.86 mm (0.70×0.70 mm median), and the slice thickness ranged from 1.25 to 10.0 mm (5 mm median). Other CT image settings were inconsistent. Of these 216 patients, 45 were excluded because of insufficient data, previous treatment, or multiple treatment, and the remaining 171 patients were enrolled in this study. The study was approved by the Wakayama Medical University Institutional review board (Approval No. 2803). All patients gave written informed consent prior to this study.

SWL procedures

SWL were performed with Lithotripter S (Dornier MedTech, Germany) under only analgesia with diclofenac sodium (50 mg). All physicians received specific and uniform technical training on SWL. All patients underwent up to a maximum 4000 shots with gradual augmentation of power at 60–90 shots per minute. Shock-wave intensity was adjusted according to physicians' judgement, based on the efficiency of lithotripsy and the degree of patient pain.

Overview of analysis methods

A flowchart of the proposed method is shown in Fig. 1. It consists of stone 3D region extraction (extraction of target stone region), followed by feature extraction (calculation of stone morphometry and multiple CT textures within the stone region), and then SVM classification (AI prediction of the SWL outcome by non-linear support vector machine (SVM) [13] with the calculated and input factors).

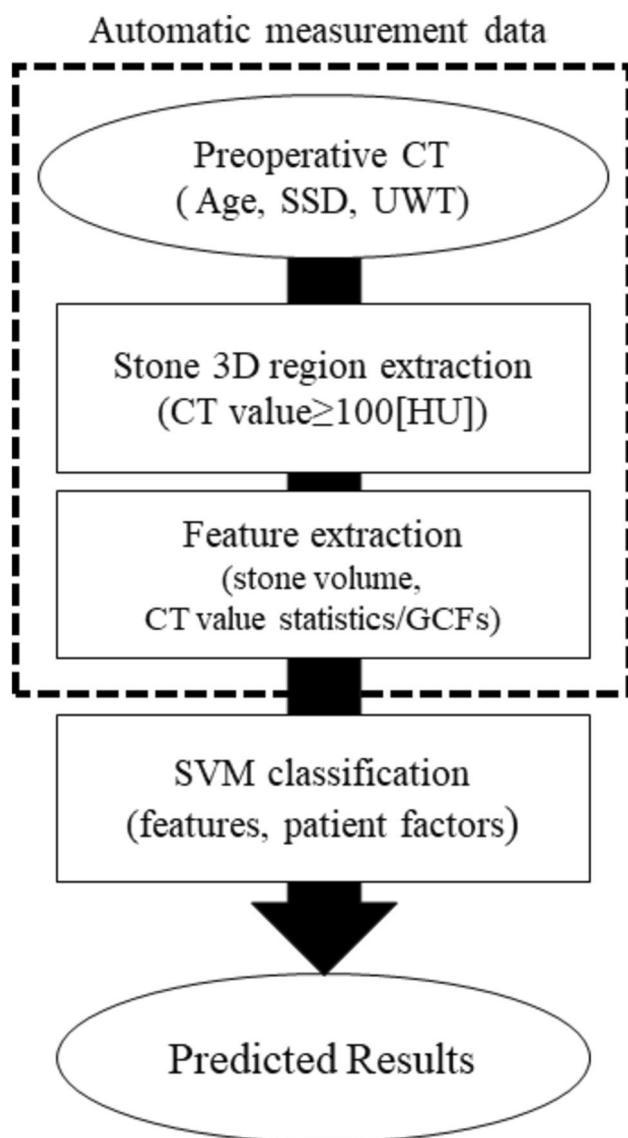


Fig. 1 Flowchart of the proposed method

Data collection

In the proposed method, 15 predictors (three non-automatic measurement factors [input data] and 12 automatic measurement factors) were used to predict SWL outcomes. Non-automatic measurement factors (input data) included patient age, SSD, and UWT, and automatic measurement factors included stone volume, seven CT value statistics (MSD, median, interquartile range [IQR], standard deviation (SD), skewness, kurtosis, and VCSD), and four gradient concentration factors (GCFs).

The measurement method of input data is shown below. SSD was calculated as the average of the distance from the center of the stone to the skin measured in three directions (0° , 45° and 90° angles). UWT was defined as the maximum

thickness of the ureteral wall at the urinary maximum stone diameter on axial images.

The automatic measurement factors are parameters that quantify the stone size or the CT image texture information inside the target stone. These factors are calculated based on the target stone area extracted by the 100 HU thresholding process. The measurement details are shown in Appendix 1.

The proposed method is specifically designed to predict the outcome of the first SWL treatment. The SWL outcome in this study was defined as cases in which complete stone-free (SF) status was obtained by a first SWL session. We evaluated patients by kidney, ureter and bladder radiographs and by ultrasonography a few weeks after the first SWL session. If residual stones required additional treatment (i.e., not fragmented at all, or the fragment being too large for spontaneous passage) which were identified, a second SWL session was performed on the same day. Complete SF status was defined as no residual stones at all according to NCCT within 3 months after the first SWL session. Patients who underwent additional SWL sessions or were transferred to another treatment (URS or PCNL) were classified as unsuccessful cases.

Predicting outcomes with artificial intelligence

AI prediction of the SWL outcome used non-linear SVM with radial basis function kernel and soft margin. The SVM was previously machine-learned using a dataset of successful and unsuccessful SWL treatments to identify the success rate of SWL treatment in target patients. Identification by the SVM was processed using a total of fifteen factors; the 12 factors calculated from CT images, and patient age, stone location, SSD and UWT.

A five-fold cross-validation (CV) method was used to evaluate the generalization performance in the proposed method. The dataset was divided into five fold: one was a test, and the others were for training the AI algorithm. Five evaluation experiments were performed with different test folds, and the generalization performance was evaluated from the average of the performance indices. The number of cases included in each fold was either 34 or 35, as shown in Table 1. The NCCT data used in this study did not have uniform resolution or slice thickness, so the data were so that the variation of slice resolution and thickness in each fold was almost equal. AUC and sensitivity at arbitrary specificity were used to evaluate the performance of predicting SWL treatment outcomes.

The contribution of the 15 factors for predicting SWL treatment outcomes was also analyzed. As in similar previous studies, the contribution to the prediction of treatment outcome was estimated from statistical significance tests between the successful and unsuccessful treatment cases. The Wilcoxon rank-sum test was employed in this study. In

Table 1 Clinical data contingency table: Resolutions are averages within each cross-validation

	<i>n</i> (SF)	<i>n</i> (Failure)	<i>n</i> (Case)	Resolution	Slice thickness	Voxel size
Fold 1	20	15	35	0.702	5.364	2.657
Fold 2	19	15	34	0.702	5.352	2.631
Fold 3	19	15	34	0.704	5.412	2.673
Fold 4	19	15	34	0.701	5.647	2.841
Fold 5	20	14	34	0.700	5.500	2.660

addition, the contribution of each factor in the multivariate prediction using SVM was also analyzed with permutation feature importance (PFI). The PFI estimates the importance of any one factor based on the degree of performance loss associated with the nullification of the factor. Disabling a factor is implemented by a random permutation of the factor.

Results

Overall, included in this study were 171 patients (118 male, 53 female). Of 171 the SWL procedures, 97 (57%) were classified as being successful. Comparison of patient characteristics and stone parameters between successful and unsuccessful cases is shown in Table 2. MSD, median, and SD were significantly lower in successful cases than in unsuccessful cases; these median values of successful and unsuccessful cases were 400.3 HU vs. 423.2 HU (MSD), 305 HU vs. 349.5 HU (median), and 250.9 HU vs. 293.5 HU (SD).

The statistical significance of the $GCF_{(0,45)}$ between the successful cases (median 0.78) and the unsuccessful cases (median 0.74) was also confirmed. Furthermore, stone volume was significantly smaller in successful cases than in unsuccessful cases (246.4 mm³ vs 310.6 mm³ on median, $p=0.021$). UWT was significantly thinner in successful cases than in unsuccessful cases (3.1 mm vs 3.9 mm on median, $p=0.023$).

The mean AUC in the prediction performance evaluation with CV was 0.742 with a 0.038 standard deviation. The highest and the lowest AUCs were 0.792 and 0.707. The ROC curve for each cross-validation is shown in Fig. 2. When the specificity was fixed at 0.733 (= 11/15), mean of the SF sensitivity in CV was 0.692 with a 0.056 standard deviation. In the same situation, the highest and lowest sensitivities were 0.737 (= 14/19) and 0.600 (= 12/20).

PFI results are shown in Fig. 3, and the graph shows the extent to which performance degrades when the factor is varied randomly. Percentage reduction from the original

Table 2 The results of Wilcoxon rank-sum test between the successful and failure SWL treatment groups

Automatic measurement	Successful (<i>n</i> = 97)	Unsuccessful (<i>n</i> = 74)	<i>p</i> value of Wilcoxon (Successful vs unsuccessful)
Possible			
CT value statistics	400.3(315.6–457.7)	423.2(376.0–510.1)	0.029
MSD, HU	305(256.9–378.5)	349.5(279.5–416)	0.035
Median, HU	375.8(283.3–488.8)	422.8(321–549.5)	0.094
IQR, HU	250.9(207.5–306.7)	293.5(211.1–360.8)	0.045
SD, HU	0.81(0.51–1.01)	0.71(0.53–0.99)	0.482
Skewness	2.51(2.03–2.92)	2.39(2.01–2.83)	0.484
Kurtosis	63.5(56.4–71.6)	65.1(57.4–69.9)	0.836
$GCF_{(0,45)}$	0.78(0.72–0.83)	0.74(0.68–0.83)	0.029
$GCF_{(45, 90)}$	0.18(0.14–0.23)	0.22(0.15–0.28)	0.059
$GCF_{(90, 135)}$	0.02(0.01–0.04)	0.02(0.01–0.04)	0.435
$GCF_{(135, 180)}$	0(0–0.003)	0(0–0.004)	0.197
Stone volume, mm ³	246.4(165.6–406.4)	310.6(207.7–486.8)	0.021
Impossible			
Age, years	60(48–69)	62(51–73)	0.203
SSD, mm	113.5(104.1–120)	112.3(99.4–123.1)	0.772
UWT, mm	3.1(2.2–4.6)	3.9(2.7–5.1)	0.023

MSD mean stone density, IQR interquartile range, SD standard deviation, VCSD variation coefficient of stone density, GCF gradient concentration factors, SSD skin-to-stone distance, UWT ureteral wall thickness Data are presented as median (quartile)

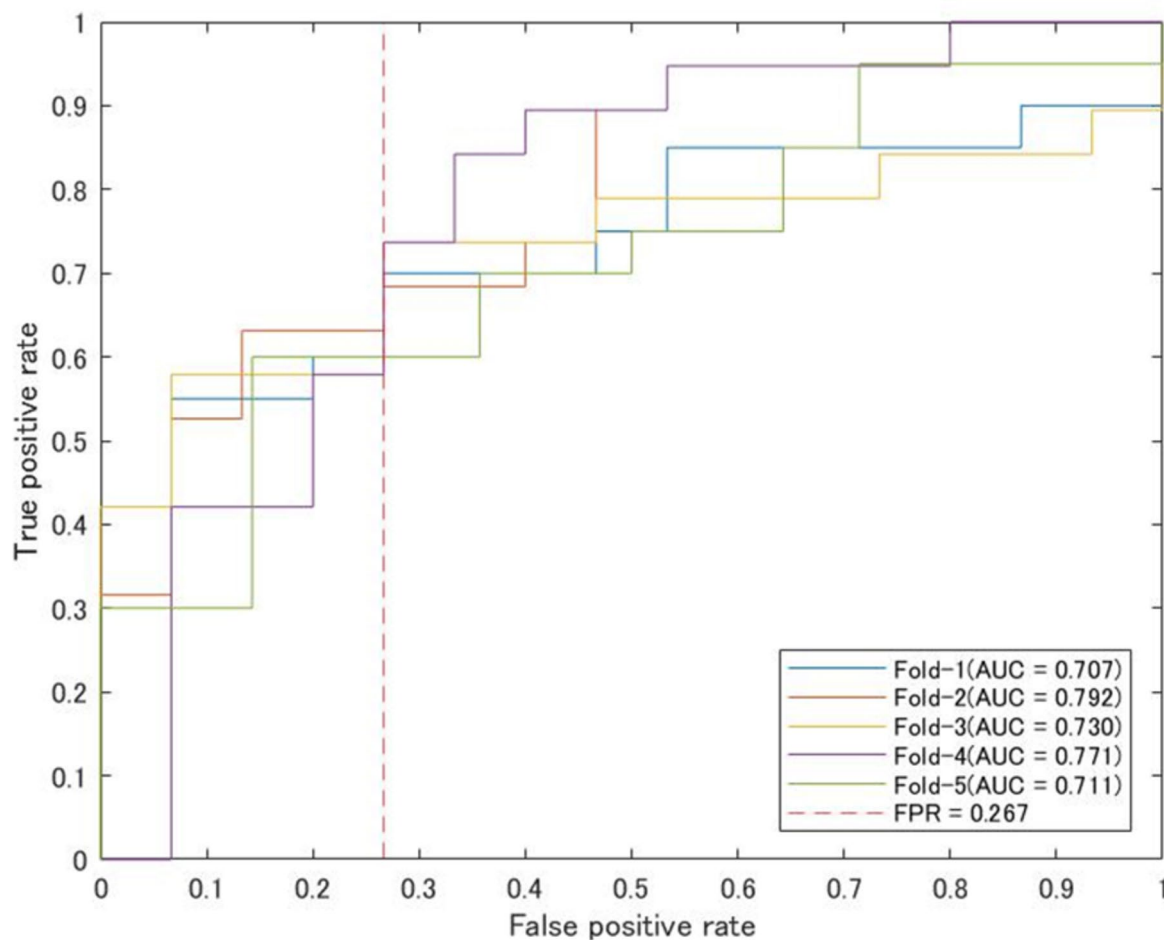


Fig. 2 ROC curve

performance is shown in descending order; the top five factors with significant reductions in prediction performance AUC due to invalidation were skewness, SSD, MSD, quartile range, and VCSD. SSD and VCSD had over 0.1 p -values in both significance tests.

Discussions

SWL, a standard treatment for urolithiasis, is still widely performed because of its low invasiveness and safety, although the SF rate of SWL is reportedly lower than that of URS or PCNL [1, 2]. To prevent renal dysfunction and urinary tract infection due to prolonged treatment, it is important to select suitable patients for SWL before treatment. For this reason, various predictors for SWL treatment outcomes obtained from NCCT have been reported, but not yet established to make an appropriate treatment choice for urolithiasis. We identified two major problems in utilizing these predictors to assist in treatment selection for urolithiasis. The first is that the inclusion of measurer

variability in the manual calculation of predictors from the NCCT image is a concern. In this study, we proposed an AI-based method for predicting SWL treatment outcomes using factors automatically measured from NCCT images. The automatic measurement is effective for avoiding inter-rater error in factors. Second, systematically predicting SWL outcomes using various factors is difficult. Several reports have predicted SWL outcomes using a single factor, but no established nomograms or formulas have combined them, and it is difficult to measure and determine all factors in the limited time available within actual clinical practice. We therefore focused on the construction of an algorithm for prediction of SWL outcomes using AI with the idea that using multiple factors for AI-based prediction will lead to results with higher accuracy.

The proposed AI prediction method for SWL treatment outcomes was applied to a retrospective dataset of 171 patients who underwent SWL for a single ureteral stone while performing validation experiments. The size of our dataset is thought to be reasonable compared with the recent clinical AI studies [14]. In the validation, the

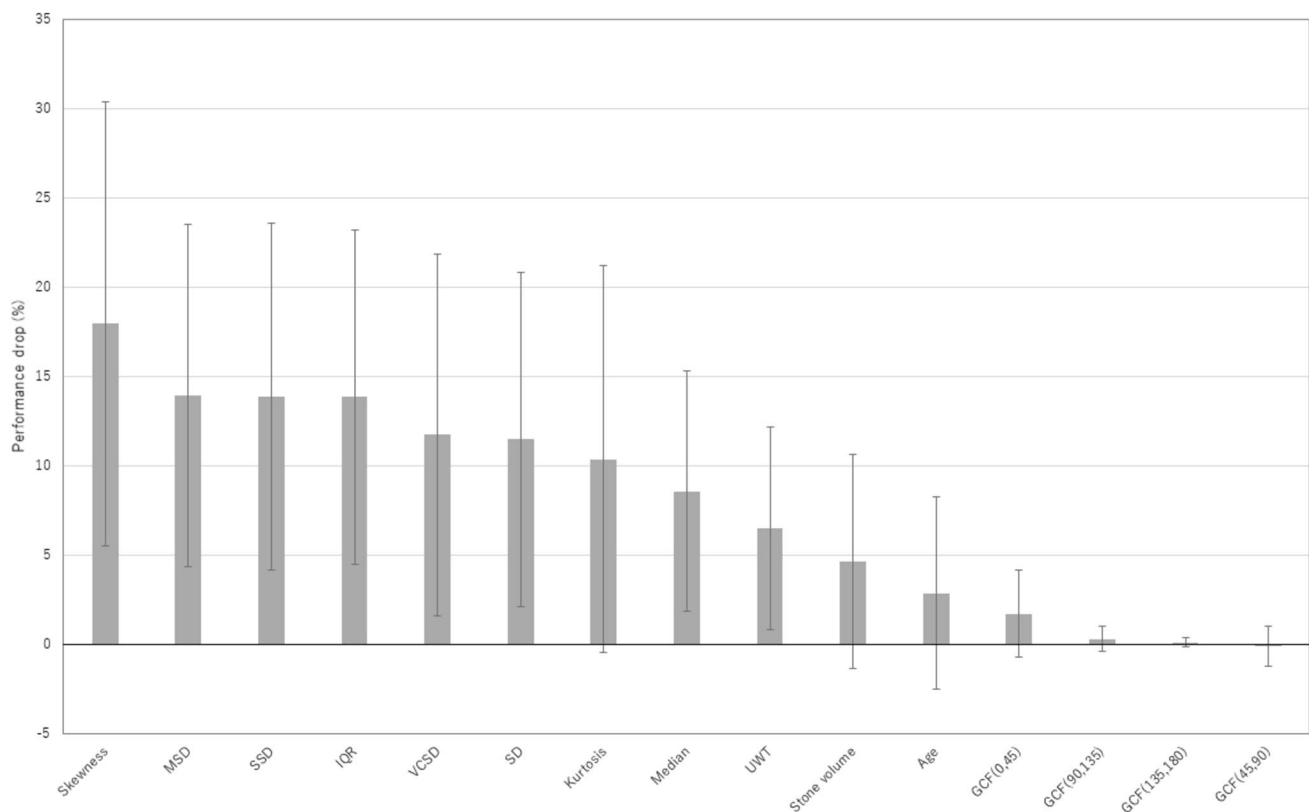


Fig. 3 Percentage reduction in discrimination performance for each factor due to PFI

proposed method successfully proceeded with all data. The dataset included CT images of various slice resolutions and thicknesses taken by multiple CT scanners. These successful results show that the proposed method is robust to data diversity, especially the thickness of the NCCT slice, and they show the feasibility of the proposed method for various clinical settings.

AUC and SF sensitivity evaluated the prediction performance of the proposed method by five-fold CV with CT data in various specifications. The data for evaluation and AI training are completely separated in the CV, so the obtained performance does not have inappropriate positive bias. The mean of the AUC was 0.742 with a 0.038 standard deviation. The mean of the SF sensitivity was 0.692 with a 0.056 standard deviation when the specificity was fixed at 0.733. Though the method proposed by Mannil et al. using a random forest classifier with few CT texture factors and the stone size showed 0.81 AUC, their study used a CT image dataset with uniform slice thickness and resolution of about 1.0 mm [11]. The effectiveness of machine learning AI generally depends on the training data characteristics and the performance of the learning algorithms [15]. The variability of the data often significantly increases the difficulty of the prediction problem, so if using the method reported by Mannil et al. it would be difficult to achieve high

prediction accuracy with the data we used in this study. Considering this point, the performance obtained in this study was thought to be comparable with the results of previous studies. The validation results obtained in this study were sufficient to prove the versatility of the proposed method.

This study analyzed the tendencies of the 15 factors in the dataset by significant tests. The results of the statistically significant difference test were generally similar to those of previous studies [16], although there were some subtle differences due to differences in the data used. The heterogeneity of CT attenuation values of stones has recently been reported to be an independent predictor of SWL outcomes [6]. Although high SD has been reported as an independent predictor for SWL success, SD was significantly higher in unsuccessful cases in this study. We attribute this to SD being strongly influenced by MSD. Statistically, the coefficient of variation in the standard deviation divided by the mean is considered to be a more accurate indicator of heterogeneity because the mean strongly influences the standard deviation. Therefore, we also used VCSD, which was defined as the variation coefficient value of CT attenuation, and because it was reported that the AUC of VCSD was significantly larger than that of SD [7]. Unfortunately, the study was limited to patients with ureteral stones, so the VCSD also did not reach significance because, compared

with renal stones, the size was smaller and the heterogeneity of CT was less likely to be reflected. In ureteral stones, the thickening of the ureteral wall at the stone site was reported to be an independent factor in predicting poor SWL outcomes [8]. The data set used in this study also showed a tendency for the UWT of the successful treatment cases to be significantly smaller than the UWT of the unsuccessful cases. Based on these findings, the dataset used in this study did not have a biased trend and was thought to be a reasonable dataset to examine, although small in size.

This study also analyzed the contribution of the 15 factors to predicting SWL treatment outcomes by PFI. The results of PFI sometimes differed from the results of the significance tests focusing on only one factor. Although SSD and VCSD of stone CT values had high *p*-values in the Wilcoxon rank-sum test, the PFI results showed that these factors had a higher contribution to the prediction by the proposed method; the random permutation of each of the two factors caused significant reductions in prediction performance AUC. These results suggest the need for variable selection that does not rely on significant difference tests when designing multivariate treatment outcome estimates.

This study has a small number of limitations. The first is that this study was retrospective, and the SWL protocol and timing of treatment effect determination were not completely consistent. However, minimum standards were established and SWL and efficacy determinations were followed. Second, CT imaging settings were not standardized. Based on these limitations, further improvement of accuracy is needed for clinical application. We will add predicting factors that were not employed in this study, as well as studies applying deep learning models, which can achieve high-performance AI. Prospective performance evaluation is also essential for actual implementation in clinical sites. We are currently preparing for a prospective study.

Conclusion

Our AI-based method for predicting SWL treatment outcomes by machine learning with a wide variety of factors, such as CT image texture parameters on the target stone, the stone size, the environment around the stone, and patient characteristic factors. The method was retrospectively validated using multiple patients' data, including CT images of various resolutions and slice thicknesses taken with scanners from different manufacturers. Also, the proposed method using multiple factors was applied to the PFI to analyze the importance of each factor for the prediction. Future validations will use a larger volume of high-resolution clinical CT image data and prospective clinical data. The application of deep learning to achieve high-performance AI is also essential future work.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00240-023-01506-7>.

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Author contributions Y.Nakamae. were responsible for all the procedures, feature extraction system construction, analysis, and writing of the manuscript. U.Kimura., and M.Nemoto. contributed for methodology creation and refinement. R.Deguchi., S.Yamashita., Y.Kohjimoto., and I.Hara. contributed project administration, clinical data collection, and lesion label annotation. All authors have read and agreed to the published version of the manuscript.

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Data availability The raw data of the results analyzed during this study are available from the corresponding author or at reasonable request. The patient dataset, including the CT image dataset, is private and not publicly available due to the protection of personal information.

Declarations

Conflict of interest All authors declare that they have no conflict of interest.

Ethical approval All procedures were carried out in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration. This study was approved by Wakayama Medical University Ethics Committee (No. 2803).

Informed consent Informed consent was obtained from all individual participants included in the study.

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